

Name: Sofia Contreras Salazar				
School: Academica Británica Cuscatleca				
Group: 4th quin...			Sex: Female	
Date of first test: 13/11/2024		Level: 8B		Age: 9:04
Date of second test: 10/10/2025		Level: 9B		Age: 10:02

Scores

No. attempted (/54)	SAS	SAS (with 90% confidence bands)										Overall ST	PR	GR (/25)	End of KS2 indicator	Progress Category
		60	70	80	90	100	110	120	130	140						
51	99											5	47	6	103	Expected

Previous & Current Scores

Date	Age at test (yrs:mths)	Level / Form	No. attempted	SAS	SAS (with 90% confidence bands)										ST	PR	End of KS2 indicator
					60	70	80	90	100	110	120	130	140				
07/05/2025	9:09	9A	49 / 54	97											5	42	102
10/10/2025	10:02	9B	51 / 54	99											5	47	103

Analysis of Curriculum content categories

Curriculum content category	Number of questions	Student % correct	Standardisation % correct	Student / standardisation difference
Number	38	68%	62%	6%
Measurement	6	25%	47%	-22%
Geometry	4	67%	40%	27%
Statistics	6	50%	63%	-13%

Analysis of Process categories

Process category	Number of questions	Student % correct	Standardisation % correct	Student / standardisation difference
Fluency in facts and procedures	13	69%	64%	5%
Fluency in conceptual understanding	22	67%	64%	3%
Problem solving	7	22%	34%	-12%
Mathematical reasoning	12	67%	57%	10%

Implications for teaching and learning

- By comparing scores from a previous administration of *PTM* it is possible to categorise progress as expected, higher or lower than expected or much higher or much lower than expected.
 - Sofia took *PTM8B* in November 2024 and from then until now has made expected progress in maths.
- Reviewing the *Analysis of Curriculum content categories* will help to identify where there are specific strengths and weaknesses and to plan next steps.
- Where performance is fairly evenly balanced across curriculum categories, this suggests that Sofia will generally demonstrate a level of understanding of mathematical concepts appropriate with this age, irrespective of whether this is in a format requiring steps in a calculation to be written down, or in mental maths when only 'jottings' or nothing is written. Fluency and agility are equally well developed in both written and mental formats and Sofia is developing the language of mathematics in line with expectations for her age group.
- Where performance across the curriculum categories are uneven, specific areas of weakness might be addressed as follows:
 - Further targeted practice in the areas identified as being relatively weaker.
 - Practical activities using equipment that is designed to help Sofia to 'see' the thinking that lies behind any concepts that are not yet secure.
 - Get Sofia to explain workings to another pupil so that any misconceptions can be highlighted and corrected through discussion.
- Sofia is generally secure in performing the basic mental calculations expected for this age group. These include age-appropriate fluency with whole numbers, fractions, decimals, percentages and ratio as well as an understanding of the number system, place value and the connections between inverse operations such as multiplication and division.
- Sofia is developing mathematical reasoning and the ability to solve a wider range of problems, including increasingly complex properties of numbers and arithmetic, and problems demanding efficient written and mental methods of calculation. Sofia has been introduced to the language of algebra to extend problem solving to a wider variety of contexts and is also developing an understanding of the geometry of increasingly complex shapes and the language needed to describe these.
- Next steps should include opportunities to stretch and develop conceptual understanding, to increase mental agility and to progress the development of formal written mathematics using age-appropriate symbols and methodologies. This can be done by emphasising the higher order skills of hypothesising or predicting (What is an approximate value for 49×11 ?); designing and comparing procedures (How many different ways can we work out $53 - 18$? Which is the most efficient?); interpreting results (If $9 \times 5 = 45$, how many other calculations can we easily work out, e.g. 0.9×5 ?) and applying reasoning (If I have 39p in my purse, what coins might they be?). In the formal written development there should be an emphasis on developing the correct use of mathematical language, for example in using ':' for ratio and the '=' sign correctly as a mathematical sentence.
- By providing practice time and frequent opportunities for Sofia to use one or more known facts to work out more facts, and by engaging Sofia in discussion and providing opportunities to explain methods and strategies to you and her peers, Sofia will be helped to make the connections between mental calculations and the written explanations. This will help to develop both problem solving abilities and the language of mathematics, thus improving both mental agility and the ability to plan and carry out accurate calculations in a more formal way.

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Analysis by question

The table below shows each question in the test and the pupil's response (correct/incorrect) compared with the group and standardisation average.

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
MM1	Number	Fluency in facts and procedures	What is six doubled?	1 / 1	68	87
MM2	Number	Fluency in facts and procedures	Multiply three by seven.	0 / 1	52	63
MM3	Number	Fluency in conceptual understanding	Write in figures the number four thousand and thirty-six.	0 / 1	36	63
MM4	Number	Mathematical reasoning	A packet of forty-five sweets is shared equally between five people. How many sweets does each person get?	0 / 1	24	64
MM5	Number	Fluency in facts and procedures	A cake is cut into thirds. How many pieces are there?	1 / 1	64	56
MM6	Measures	Fluency in conceptual understanding	My watch says half past two in the afternoon. What would a twenty four hour digital watch say?	0 / 1	4	28
MM7	Number	Mathematical reasoning	One car has four tyres. How many tyres do three cars have?	1 / 1	72	80
MM8	Number	Mathematical reasoning	There are six eggs in one box. How many eggs are there in seven boxes?	0 / 1	8	52
MM9	Number	Fluency in conceptual understanding	What is twenty-five take away thirteen?	1 / 1	64	66
MM10	Number	Fluency in facts and procedures	Write a multiple of six that is between fifteen and twenty-five.	0 / 1	16	44
MM11	Number	Fluency in conceptual understanding	Look at these numbers. Write down all the odd numbers.	1 / 1	52	64
MM12	Number	Fluency in facts and procedures	Add these numbers: forty-one, twenty-five and fourteen.	0 / 1	32	43
MM13	Number	Mathematical reasoning	There are thirty-two people in a restaurant. Thirteen are adults. How many are children?	1 / 1	28	36
MM14	Measures	Mathematical reasoning	I have twenty-seven pence. I want to spend thirty-five pence. How much more money do I need?	1 / 1	60	61
MM15	Measures	Fluency in facts and procedures	How many days are there in eight weeks?	0 / 1	16	47
AU1	Number	Fluency in conceptual understanding	Draw lines to connect the matching fractions.	2 / 2	86	89
AU2	Number	Fluency in conceptual understanding	What fraction of this shape is shaded?	1 / 1	36	78
AU3a	Number	Fluency in conceptual understanding	Which day had the lowest midnight temperature?	1 / 1	56	59

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
AU3b	Number	Fluency in conceptual understanding	Which day had a midnight temperature 1 degree higher than Friday?	1 / 1	40	50
AU4	Number	Mathematical reasoning	On Sunday the midnight temperature was -2 degrees. The next day, the midnight temperature was 5 degrees higher. What was it?	1 / 1	48	49
AU5	Geometry	Fluency in facts and procedures	Name the 3 shapes that are labelled.	1 / 1	28	46
AU6a	Geometry	Fluency in facts and procedures	In which labelled shape are all the angles right angles?	1 / 1	48	60
AU6b	Geometry	Fluency in facts and procedures	In which labelled shape are all the angles greater than a right angle?	1 / 1	32	42
AU7	Number	Mathematical reasoning	Here are 3 more fraction sums that make 1 whole. Fill in the missing numbers.	2 / 2	22	62
AU8	Measures	Mathematical reasoning	Write the times in the correct boxes.	0 / 2	44	36
AU9	Statistics	Fluency in conceptual understanding	Put a circle round the country that has the least tigers.	1 / 1	64	83
AU10	Statistics	Fluency in conceptual understanding	Roughly how many tigers are there in Malaysia?	1 / 1	76	81
AU11	Statistics	Fluency in conceptual understanding	Put a circle round the 3 countries that have less than 500 tigers.	1 / 1	68	70
AU12a	Number	Fluency in conceptual understanding	Which number has 7 as its hundreds digit?	0 / 1	60	83
AU12b	Number	Fluency in conceptual understanding	Which number is 10 times bigger than 62?	1 / 1	40	63
AU12c	Number	Fluency in conceptual understanding	Which number is not a whole number?	0 / 1	60	70
AU12d	Number	Fluency in conceptual understanding	Which is the smallest number?	1 / 1	92	81
AU13	Number	Fluency in facts and procedures	Here is a number machine. Fill in the missing number.	1 / 1	84	74
AU14a	Number	Mathematical reasoning	Fill in the missing number.	1 / 1	68	66
AU14b	Number	Problem solving	What number comes out the same as it goes in?	0 / 1	28	40
AU15a	Number	Fluency in facts and procedures	Which is the biggest number?	1 / 1	88	94
AU15b	Number	Fluency in facts and procedures	Write in a number that is bigger than the biggest number here.	1 / 1	92	86
AU16a	Number	Fluency in facts and procedures	Which are the smallest two numbers in the list?	1 / 1	100	93
AU16b	Number	Mathematical reasoning	Write in any whole number that is between the 2 smallest numbers.	1 / 1	64	74
AU17	Measures	Fluency in conceptual understanding	Which two of these are equal to seventy pence?	1 / 2	54	68
AU18	Number	Fluency in conceptual understanding	What number is exactly between 41 and 55?	1 / 1	36	31
AU19	Number	Fluency in conceptual understanding	What number is exactly between 4.5 and 5.5?	1 / 1	48	51
AU20	Number	Fluency in conceptual understanding	What number is exactly halfway between $\frac{2}{7}$ and $\frac{4}{7}$?	1 / 1	56	71

Question number	Curriculum category	Process category	Question content	Score (/x)	Group % correct	Standardisation % correct
AU21	Geometry	Problem solving	Write the letters of each shape into the correct boxes.	1 / 3	28	31
AU22	Number	Mathematical reasoning	Now choose 4 different numbers to show more multiplications with the answer 32.	1 / 2	36	47
AU23	Measures	Fluency in conceptual understanding	At the bank, Mr Smith changes a £10 note for 20p pieces. How many 20p pieces does he get?	0 / 1	16	29
AU24	Number	Mathematical reasoning	What is the smallest possible number you can make using all 4 of these cards?	1 / 1	76	84
AU25	Number	Problem solving	Arrange the cards to make 2 numbers whose sum is 74.	1 / 1	48	57
AU26	Number	Problem solving	Arrange the cards to make 2 numbers whose difference is 26.	0 / 1	32	35
AU27a	Number	Problem solving	How many stickers will be left when she has made all the cards?	0 / 1	16	21
AU27b	Number	Problem solving	How many more invitations can she make with the left-over stickers?	0 / 1	8	24
AU28	Statistics	Fluency in conceptual understanding	How many babies were at the medical centre on that Wednesday?	0 / 1	20	40
AU29	Statistics	Fluency in conceptual understanding	Put an x on the graph in question 17 to show John's heartbeat.	0 / 1	56	70
AU30	Statistics	Problem solving	Which 2 of these age groups could Erin be in?	0 / 1	8	33